

Downloading Vintages from the ECB Real Time Database

Experience Report and Construction of the Vintages

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Executive Summary

Every vintage from January 2001 to September 2026 (261 vintages) can be obtained for all 278 series by following the [RTD page on the ECB Data Portal](#). The instructions for 2001–2014 are clear and accurate, but for later vintages they are not. For the later vintages, the page provides a single example for reference, and several statements on the page are wrong: the update schedule, the description of the pre-2001 files, and the coverage of the web service, which actually starts in January 2001, not 2015. **The main difficulty is that the address returns a log of revisions rather than vintages, and neither the page nor the API help explains how to rebuild a vintage from it.** One series can be downloaded without programming, by pasting the page’s address into a browser; the full set of 278 series takes 278 slow requests and is practical only with a script.

All 278 series were downloaded and every vintage was rebuilt (Section 3). For 2001–2014 the rebuilt vintages are identical to the ECB’s own vintage files wherever those files have data, and the later vintages that were compared with Eurostat agree with it (Section 4). **The vintages are all there, but many series are not kept up to date. In the latest vintage, 107 of the 278 series end at least ten months earlier than they normally would, and since 2015 a total of 144 series have been behind or without observations for two or more vintages in a row (box 7).** See Section 6 for a list of suggestions for the ECB.

1 Website Claims vs. Findings

Topic	Website Says	Found
Update schedule	“updated semi-annually, at the beginning of January and July”	False. The page itself shows “Dataset last update 9 September 2026”, and that vintage was online by 3 October. The meeting of 23 July 2026 has no vintage.
Freezing	“always takes place on a working day ahead of the Governing Council meeting at precise time 3.30 p.m.”	Mostly true. Every row is stamped 15:30, and 248 of the 261 dates are the working day before a monetary policy decision.

Topic	Website Says	Found
Before 2001	A zip file of “38 variables containing data vintages prior to January 2001”, in CSV	False. Only one, producer prices, has vintages before January 2001 (from October 1999); none goes beyond June 2009. The read-me inside the zip refers to a file of cut-off dates, Cut-off.xls , that is not in the zip. And the files in the zip are Excel, not CSV.
2001–2014	Three zip files (monthly, quarterly, annual)	True. Each holds 168 CSV files, one per Monthly Bulletin, each a full snapshot.
From 2015	“can be extracted using ECB Data Portal web services”	True. This works for one series per request. Also, without includeHistory=true the address returns only the latest vintage. It also goes further back than the page says: January 2001 onwards.
Technical documentation	“recommended to become familiar with the technical documentation, prior to proceeding with the data download”	Unclear. Names each zip-file column by letter, but gives no web-service keys and does not mention includeHistory or the columns of the download.

2 Easy Mistakes for Users

1. The download is a log of revisions, not a set of vintages.

The page says its example address “can be adapted to extract the history for any RTD series.” The file it returns is not a copy of the series at each vintage but a list of changes: a row appears only when a value is published, revised or deleted. In the page’s example series, for instance, the value for 2024 is part of 10 vintages but appears only once in the file, because it has not changed since it was first published. Each row says what happened ([ACTION: Replace](#) or [Delete](#)) and when ([VALID_FROM](#) or [VALID_TO](#)).

To rebuild the series as it stood on a given date, take, for each period, the most recent row dated on or before that date. If that row is a [Delete](#), the period was not in the series on that date. Neither the page nor any of the nine API help pages explains this, and none of them mentions these three columns.

2. Revisions are recorded as a deletion followed by a new value.

On 20 January and 8 September 2021, many series were revised (212 and 189 of the 278). The file records these revisions in two steps: it first deletes the old observations (stamped `15:30:00`) and then enters the revised values one second later (`15:30:01`). For real GDP, every quarter was handled this way, and every value changed.

Someone who looks only at the date sees the deletion and the new value on the same day. Because deletions are listed at the end of the file, the deletion looks like the last word, and the series wrongly disappears from those two vintages. Comparing the full time stamp, not just the date, avoids it.

3. Requests for many series at once lose the deletions.

Each series is requested by a key with five parts, such as frequency, country and variable. The API lets a user leave parts blank to mean “any” (it calls this “wildcarding”): filling in only the first part as quarterly, for example, returns every quarterly series in one request. This is much faster than one request per series, but the file leaves out every record of a period being removed, without saying so. The data then look as if nothing was ever removed, and values the ECB withdrew still appear in later vintages.

For example, real GDP for 1991–1994 was removed on 30 November 2005. Requested on its own, real GDP shows that removal, and the vintage of 30 November 2005 starts in 1995. Requested with a wildcard, the removal is missing, and the same vintage still starts in 1991.

The fix is to name every series in full. Requesting each series on its own returns the removals; joining two full keys with `+` also did, in a test with real and nominal GDP.

4. The old files and the web service name the series differently.

The 2001–2014 vintages come as three zip files (monthly, quarterly and annual), each holding one CSV file per month, with one column per series. The columns are headed by the series codes in use that month, which change over time for 225 of the 268 series, and none of them is the name the web service uses. Real GDP, for example, is column BM in the quarterly files but `Q.Sø.S.G_GDPM_TO_C.E` in the web service. The technical documentation lists what each column contains, but not the web-service name.

The web service does say which column each series is in, but neither the page, the API help section nor the technical documentation tells the user where to look. Each series’ description has a `COMPILATION` field ending with the column letter, such as “series: BM”. For 266 of the 268 series that have one, the letter is right (Section 4). The two M3 annual growth series have their letters swapped, and the other 10 series have none, so I matched them by comparing their data with the columns no other series had claimed, which works, but is not an ideal solution.

5. Dates.

Each vintage is dated by the day its data were frozen, not by the day of the meeting it was prepared for. Usually that is the working day before a monetary policy decision, so the vintage for the decision of 10 September 2026 is dated 9 September. This holds for 248 of the 261 vintages, including every one since 2015. The other 13, all from 2001 to 2013, were frozen two days before a decision, on the decision day itself, or in mid-January after a decision. Because the page's footnote points to the decision dates, a user looking for a vintage dated on a decision day will usually not find one.

The zip files are named by Monthly Bulletin month: 14 of the 168 hold a cut-off from the previous month (200102 holds 31 January 2001).

6. Downloading the full set is slow and unreliable.

A user who wants every series has to make 278 requests, one per series, and each one is slow. A request often takes one to three minutes, and some fail with a time-out ("504") and have to be repeated.

Worse, some files arrive cut off part-way through while the server reports success. This happened to 13 of the 278 series, some of them on several tries in a row, and the cause was not found. Because the `Delete` rows come last, a cut file loses them first, and nothing warns the user. To be safe, a user has to check that every file ends with a complete row and request it again if not, which makes a full download hard work for anyone and impractical without a script.

7. Many series are not kept up to date.

Every series has a vintage for every date, but that does not mean the series was up to date on that date. A series is called *behind* here when its last observation is older than is usual for that series by at least three months (monthly series), two quarters (quarterly) or one year (annual).

In the vintage of 9 September 2026, 107 of the 278 series are behind. Of these, 72 were last updated on 4 June 2025, among them most of the money, survey and balance of payments series, and 18 on 29 October 2025 (industrial production, producer prices and retail sales). Industrial production, for example, ends in August 2025 in that vintage, while Eurostat's figures run to July 2026.

Earlier vintages are affected as well, including for series that are up to date today.

- Nine series have no observations at all in any vintage from January 2015 to March 2024. They are five government bond yields, two raw material price series and two producer price series.
- Twenty-one survey series have no observations in the four vintages from February to June 2023.
- The 10 euro area unemployment series, the 14 employment series, 6 exchange rate series and construction prices were behind for between 8 and 23 vintages in a row, in each case up to the vintage of 4 June 2025. Unemployment, for example, ends in January 2024

in every vintage from March 2024 to June 2025.

In all, 144 of the 278 series were behind or had no observations in at least two consecutive vintages since 2015. The number affected is 9 in every vintage from 2015 to 2020, between 21 and 76 in the vintages of 2023, and at least 100 in every vintage of 2026.

The ECB's live database ends in the same period as the rebuilt vintage for every one of the 278 series, so the gaps are in the ECB's data and not in the rebuilding. Nothing in the download marks them, because a series that is not updated simply has no new rows. A user who treats a vintage as the data available on that date will, for these series, be using figures that were many months old, or none at all. Two files list the series and vintages affected (Section 5).

3 How the Vintages Were Built

All 278 series were downloaded on 3 October 2026, and every vintage from January 2001 to September 2026 was rebuilt from them, in three steps.

Step 1: Downloading the logs

Each series was requested on its own, by its full five-part key, from the address

```
https://data-api.ecb.europa.eu/service/data/RTD/<key>?includeHistory=true&format=csvdata&detail=dataonly
```

The option `includeHistory=true` asks for every version of the series rather than the latest one only; `format=csvdata` gives a CSV file; and `detail=dataonly` leaves out the columns of descriptive attributes, which are not needed to rebuild the vintages.

Each series was requested by its full key because requests that leave part of the key blank leave out every row recording a removal (box 3). Each file was also downloaded twice and kept only if the two copies were identical and every row was complete, because the server sometimes cuts a file short while reporting success (box 6).

The 278 files hold 4,131,911 rows recording a published value and 75,848 recording the removal of a period; 277 of the 278 series have at least one removal.

Step 2: Rebuilding each vintage

Each row of a log refers to one period of the series and records one event. A row marked `Repl` records that a value was published for that period, and its time stamp (`VALID_FROM`) says when. A row marked `Delete` records that the period was removed from the series, and its time stamp (`VALID_TO`) says when. Every time stamp in the logs is 15:30:00 or 15:30:01 on the day the data were frozen, so each row was dated by the day alone.

The vintage dates are all the days that appear in any of the 278 logs. There are 261, twelve a year from 2001 to 2014, eight a year from 2015 to 2025 and five so far in 2026. The rule for the vintage of a given date is then the following. For each period, take the latest row dated on or before that

date. If it is a **Repl**ace row, its value is the value of the period in that vintage. If it is a **Dele**te row, or if there is no row yet, the period is not part of that vintage. A series that did not change on a given date therefore carries its previous values into that vintage, which is exactly how it stood on that day.

One case needs care. In two vintages, 20 January and 8 September 2021, the removal of the old observations and the entry of the revised values fall on the same day (box 2). The rule therefore lets the **Repl**ace row win when both fall on the same day. This was checked across every log from 2001 to 2026. A period has both kinds of row on one day 34,989 times, all of them on those two dates, and in every case the removal is stamped before the new value. No period ever has two values published on the same day. The rule therefore always reproduces the order in which the ECB recorded events.

Step 3: Writing the files

Each vintage of each series was written to its own Excel workbook, in the layout used by ALFRED, the St. Louis Fed's archive of vintages. A workbook covers the periods from the first to the last with a value, and a period inside that range with no value is left blank. The short names (such as "Real GDP" or "Unemployment rate, women") were translated from the concept code that forms the fourth part of each key, and the ECB's own title and description are kept in each workbook so that every name can be checked.

The workbooks were checked against the rebuilt vintages they were written from. For every series, both the latest vintage and one vintage drawn at random were opened, 555 workbooks in all, and every date and every one of their 132,120 values agrees to the 15th significant digit.

4 Was My Rebuild of the Vintages Correct?

For 2001–2014 the same vintages exist in two places. The zip files store each vintage exactly as it was, while the web service (API) only gives the list of changes, from which each vintage has to be rebuilt. If the rebuilt vintages match the zip files, then the web service is complete for those years and the rebuilding script is correct, which gives good reason to trust it for later vintages as well. So for every month from January 2001 to December 2014, each series was rebuilt from the web-service download as it stood that month, and checked against that month's zip file to see whether it has the same periods and the same numbers.

Matching the two sources required two links. The first is between series, and is described in box 4: 266 of the column letters given by the web service are right, two are swapped, and 10 had to be found by comparing the data. The second is between dates. The zip files are named by the month of the Monthly Bulletin they fed, not by the day the data were frozen (box 5). Each rebuilt vintage was therefore assigned to the month of the date one week after its freeze, and each zip file was compared with the latest vintage assigned to its month or an earlier one. A comparison counts as a match only if the periods are the same and every value is identical.

Answer: Yes

Every one of the 278 series matches its zip-file column in every month in which that column has data. That is 42,132 comparisons of a series with one month's file, covering 6,616,172 values, all identical, with no tolerance for rounding.

In another 4,572 comparisons the zip-file column is empty. In 4,276 of them the rebuilt vintage is empty too. In the other 296, which involve 20 series, the zip file does not carry the series while the web service still holds earlier values for it. The main case is the five government bond yields and the two raw material price series, which are missing from the zip files from 2008 but stay in the web service until April 2011. So wherever the zip files have data, the web service gives the same data, and for those months the zip files are not needed.

The zip files do hold 44 series that the web service does not have at all, 29 of them for the United States or Japan. Anyone who needs those still has to use the zip files.

The vintages from 2015 have no such files to be compared with, so they were checked against Eurostat instead. In the latest vintage, nine quarterly series equal Eurostat's figures for the 21-country euro area in every quarter since 1995. They are GDP and private consumption, each in real and nominal terms, real government consumption, investment, exports and imports, and employment. And in six vintages from 2020 and 2026, the newest quarter of real GDP shows the growth rate in the Eurostat release published just before the vintage date. The vintage of 3 June 2020, for example, has the first estimate for the first quarter of 2020, a fall of 3.8%, and the vintage of 15 July 2020 has the revised fall of 3.6% that Eurostat published on 9 June.

These checks show that the vintages reproduce what the ECB published. They do not show that every series is up to date, and box 7 shows that many are not.

5 The Files

The vintages, the downloads they were built from and the code accompany this report.

Folder	Contents
data/processed/vintages	One workbook per variable per vintage, 67,532 in all, in 24 folders named by indicator (GDP, Unemployment, Consumer prices and so on) and one sub-folder per variable. The file <code>GDP/Real GDP/vintage_2008-06-04.xlsx</code> , for example, is euro area real GDP as the ECB had it on 4 June 2008. The list of all 278 variables, with the ECB's title and series key for each, is in <code>_variable_list.csv</code> .
data/processed/raw_vintages	The same data as one table per series, with periods in rows and the 261 vintage dates in columns. This form is the easier one for studying revisions.
data/raw/history	The 278 logs as downloaded from the ECB on 3 October 2026, unchanged. Next to them are the list of series, the ECB's code lists and, in <code>old_files</code> , the ECB's zip files for 2001–2014.
code	The four Python scripts that download the logs, rebuild the vintages, check them against the 2001–2014 files and write the workbooks.
output	The result of the check against the 2001–2014 files, one row per series and month. Also two lists for box 7: <code>series_last_update.csv</code> gives the last update of each series, and <code>series_gaps.csv</code> gives, for each series, the vintages in which it is behind or has no observations.

Each workbook has two sheets, as an ALFRED download does. The first describes the series: a short name, the ECB's own title and description, the series key, units, frequency, seasonal adjustment and the vintage date. The second has two columns, the date each value refers to (the first day of its month, quarter or year) and the value itself. A variable has a workbook only for the vintages in which it has observations. Of the 278 series, 228 start in 2001, 26 in 2002 or 2003 and 24 in 2012.

Three further points concern particular series.

- **Taxes less subsidies on products, at current prices** (`RTD.Q.S0.S.G.TLSP_TO_U.E`). From the vintage of 5 November 2014 to that of 4 June 2025 this series is identical to nominal GDP, and in later vintages it is within 2% of nominal GDP. Earlier vintages are on the scale of taxes less subsidies. A note to this effect is in its workbooks.
- **Two Euro Stoxx sector indices** carry conflicting labels in the ECB's metadata. The ECB's concept names say cyclical and non-cyclical goods, while its series titles say consumer staples and consumer discretionary. Both labels are recorded in the workbooks.
- **Composition and definitions change over time.** The euro area series follow the changing membership of the euro area, and some definitions changed between vintages. The vintages show what the ECB published at the time, including these changes.

The scripts need Python 3 with the `pandas` and `xlsxwriter` packages and are run in order from the main folder. The first is slow, because the server often takes one to three minutes to answer a request for a series it has not been asked for recently. Running it again later picks up any vintages the ECB has added since 3 October 2026.

<code>python code/01_download_rtd_history.py</code>	downloads the 278 logs
<code>python code/02_build_vintages.py</code>	rebuilds the vintages
<code>python code/03_check_against_old_files.py</code>	compares them with the 2001–2014 files
<code>python code/04_make_readable_vintages.py</code>	writes the workbooks

6 Suggestions

1. Explain on the page that the download is a list of changes, not a set of vintages, and how to rebuild a vintage from it, including how revisions show up as a deletion followed by a new value (boxes 1 and 2 of Section 2).
2. Better still, post one file per vintage from 2015, as for 2001–2014, so that users need not rebuild anything from the logs.
3. Say on the page which series are no longer updated and since when, or resume updating them (box 7).
4. Make requests for many series at once include the periods the ECB removed. At present such requests leave them out without warning, so users end up with values the ECB has withdrawn (box 3).
5. Make the server report an error when a file is cut off, instead of reporting success.
6. State that the address covers January 2001 onwards, correct the stated update schedule, and explain why the meeting of 23 July 2026 has no vintage.
7. Say on the page that each vintage is dated by the day its data were frozen, not by the decision date.
8. Add each series' web-service name to the technical documentation, fill in the column letter for the 10 series that lack it, and correct the two swapped M3 letters.
9. Correct the page's description of the pre-2001 zip file, which holds Excel files and only one series going back before 2001, and add the missing `Cut-off.xls`.